MUSTHOFA JOKO ANGGORO

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RESEARCH INTERESTS

Edge AI systems, resource-aware computing in edge-fog-cloud continuum, sustainable computing, energy-efficient systems, distributed systems architectures, hardware-software co-design

EDUCATION

University of Indonesia (UI) — Jakarta, Indonesia
Bachelor of Computer Science, Information Systems — *Expected Graduation: Aug 2026*

- **GPA: 3.84/4.00** (8th Semester, Dean's List)
- **Thesis:** Implementation of Morris Mano Computer Architecture with Enhancements on Xilinx Spartan-3 FPGA (X3700AN) using VHDL — demonstrating hardware design, digital logic optimization, and system-level architecture implementation
- **Relevant Coursework:** Computer Architecture, Operating Systems, Data Structures & Algorithms, Introduction to AI & Data Science, Statistics & Probability, Data Communication Networks, Database Systems

Nanyang Technological University (NTU) & ITS — Singapore & Surabaya
INSPIRASI–NTU Summer Program 2025 (LPDP Scholarship Awardee) — *July 2025*

- Conducted sustainability-focused research on smart port automation and green technology adoption in distributed maritime systems, analyzing IoT sensor networks, automation architectures, and energy-efficient deployments
- Produced comprehensive technical report proposing tiered port automation framework emphasizing renewable energy utilization and carbon emission reduction — directly aligned with sustainability objectives in edge computing systems
- Collaborated with international research teams on applying systems thinking to environmental challenges

SYSTEMS & RESEARCH EXPERIENCE

Software Engineer Intern — Traveloka (Travel Technology Company)
Systems Engineering & Backend Development — *Aug 2025 - Present*

- **Edge AI-Relevant Skills:** Resource optimization, distributed system architecture, fault tolerance, cloud infrastructure (AWS, Terraform), containerization (Docker, Kubernetes), performance tuning, reliability engineering

TECHNICAL SKILLS

Systems Programming & Languages: C, Python, Java, VHDL (hardware description), Go, TypeScript
Hardware Design & Architecture: FPGA development (Xilinx Spartan-3), VHDL, digital logic design, computer architecture implementation, hardware-software co-design
Machine Learning & AI: Scikit-learn, TensorFlow basics, supervised/unsupervised learning, data preprocessing, model evaluation, imbalanced learning techniques
Systems & Infrastructure: Microservices architecture, distributed systems, containerization (Docker, Kubernetes), cloud infrastructure (AWS, Terraform), CI/CD pipelines
Databases & Optimization: PostgreSQL, MongoDB, SQL query optimization, performance tuning, database trigger design
Operating Systems & Low-Level: Linux, Unix, Windows — OS internals, process management, memory management, system-level programming

MACHINE LEARNING PROJECTS

Used Car Sales Prediction — Course Project - AI & Data Science
Machine Learning Analyst — *Apr 2025 - May 2025*

- Built comprehensive predictive modeling system for used car sales using imbalanced classification, regression, and clustering techniques to derive actionable business insights from complex datasets
- Implemented multiple ML algorithms (Gradient Boosting, Random Forest, Logistic Regression, Linear Regression, Naive Bayes, K-Means) with emphasis on handling imbalanced data through oversampling and undersampling techniques
- Conducted extensive exploratory data analysis (EDA) and feature engineering, achieving robust model performance through systematic preprocessing, evaluation, and optimization using Scikit-learn
- **Tools & Techniques:** Python, Scikit-learn, Pandas, NumPy, imbalanced learning methods, model evaluation metrics

Additional Projects: Database-driven web systems (Django, PostgreSQL with triggers/stored procedures), microservices architecture (Spring Boot), enterprise system development with clean architecture principles